

PREDICTING PROPOSED LEGISLATION SUCCESS

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Introduction/Abstract

The United States' Congress proposes thousands of bills every year that are the basis for the legal system of the country. However, less than 10 percent of bills on average end up passing into law. The goal of our project was to use the text that makes up the bills with Bayesian Modeling and Convolutional Neural Networks (CNNs) to predict whether a bill will become law. Lawmakers could use these models before proposing a bill to get a better understanding of it's likelihood of passing, allowing them to refine it or choose to abandon it. The aim is to optimize not just for overall accuracy but also for accuracy of the predictions of "Became Law" status. The Bayesian Model, while for general purpose classification would not be entirely useful, does a good job optimizing for the "Became Law" category. Similarly, the binary classification CNN performs better than the multi-label classification CNN, making both models useful for predicting the "Became Law" status of a proposed bill.

Data Source

Our data was obtained from the Congress.gov site. The URL for each bill was obtained through their API and were used to get the PDFs of each bill, where the text was exported from. The legislation webpage which contains information such as sponsors, committees, and latest status of proposed and accepted bills. The possible statuses include 'Introduced', 'Failed Senate', 'Failed House', 'Passed House', 'Passed Senate', 'Resolving Differences', 'Vetoed by President', 'Became Law' and 'Became Private Law'. It is important to note the severe data imbalance here due to the very low rate of bill passage, which was accounted for in the model designs.

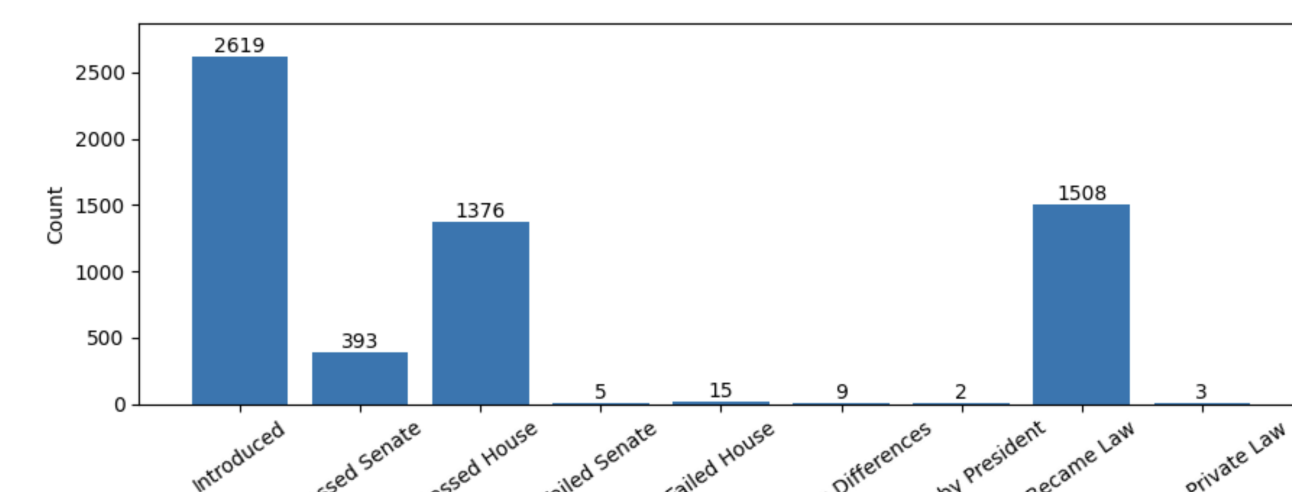


Fig. 1: Bill Status Distribution

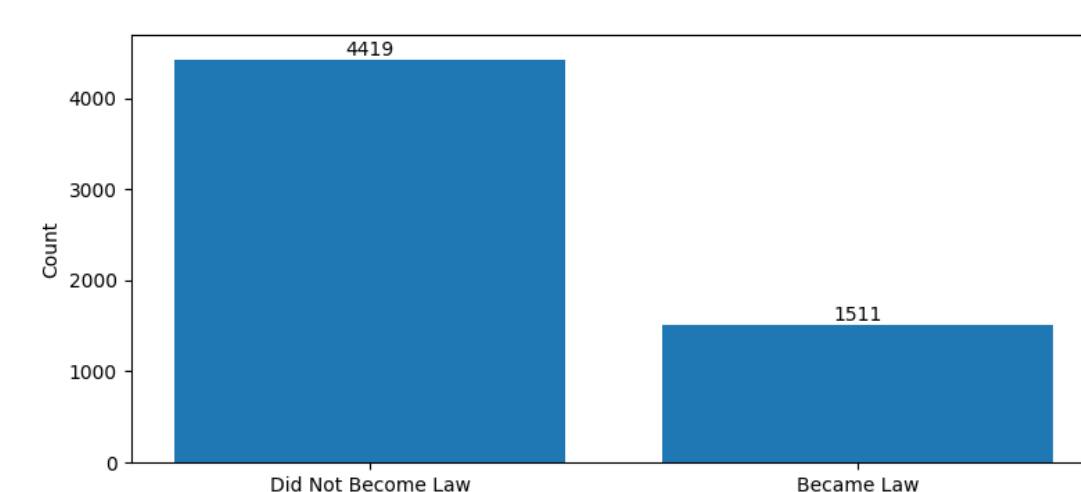


Fig. 2: Binary Bill Status Distribution

Methodology: Bayesian Approach

Using the Bayesian approach to classify Bills relies on setting up the likelihood function, prior estimates, and marginal distribution to estimate the conditional likelihood.

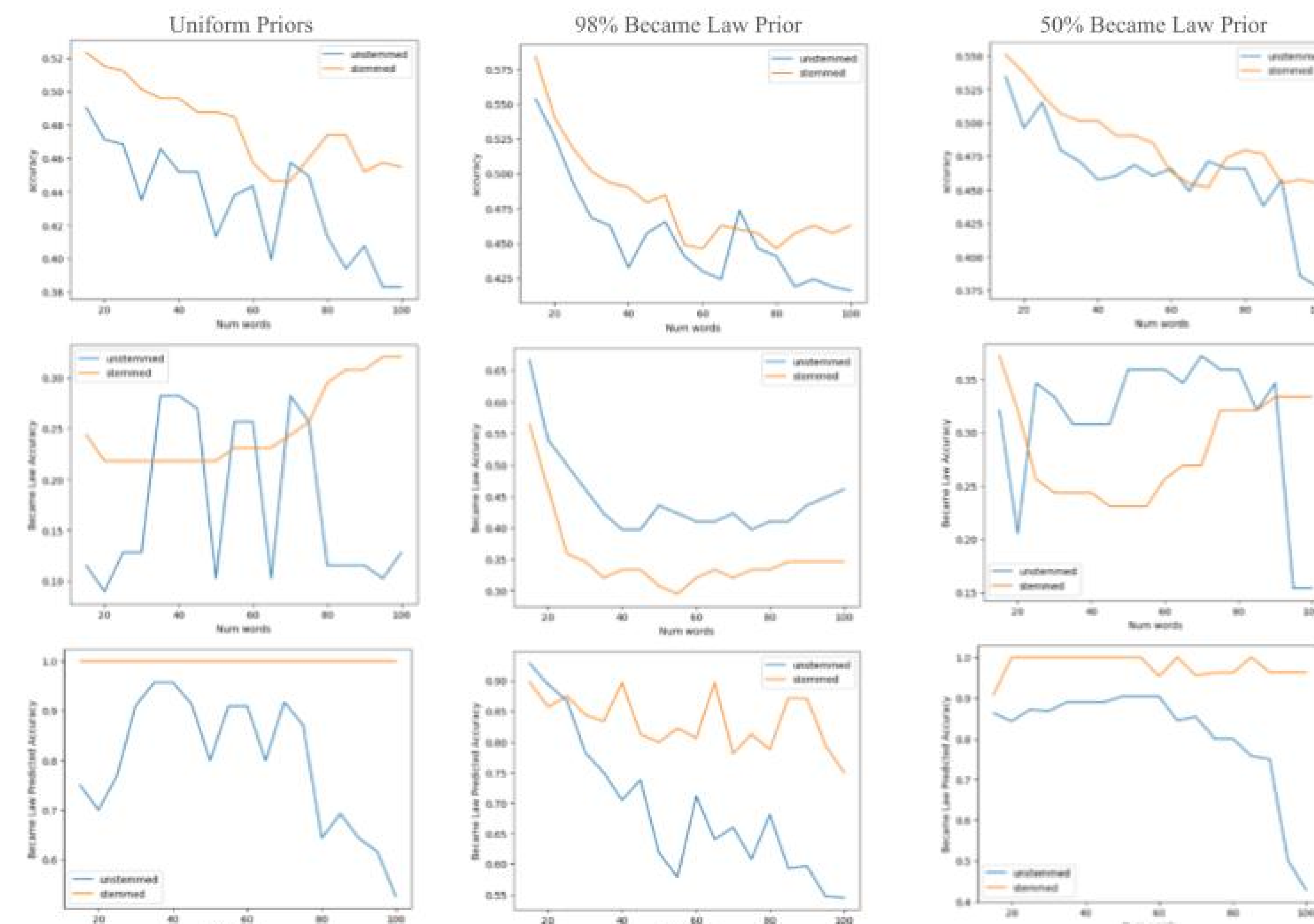
$$p(o|t) = \frac{p(t|o)p(o)}{p(t)}, p(t|o) = \prod_{i=1}^n p(w_i|o_j) * a_i + [1 - p(w_i|o_j)] * (1 - a_i)$$

o : outcome of the bill (Became Law, Introduced, etc), t : text in bill, n : number of top words used, w_i : i th word, a_i : if i th word appears in text

First the likelihood, $p(w_i|o)$, is calculated using the training dataset by taking the top n words of the entire data set and finding the distribution of their occurrences for each outcome. Then, to calculate the posterior, $p(o|t)$, the product of all of probabilities for a particular outcome is calculated to find the likelihood, $p(t|o)$, is multiplied by the prior, $p(o)$, and divided by the sum of all of the $p(t|o)p(o)$, which is the marginal, $p(t)$. The goal was to maintain overall accuracy, Became Law outcome accuracy, and Became Law prediction accuracy. Number of words between 5 and 100 and 5 separate priors were all tested on validation data to produce the optimal word vector and prior for final test.

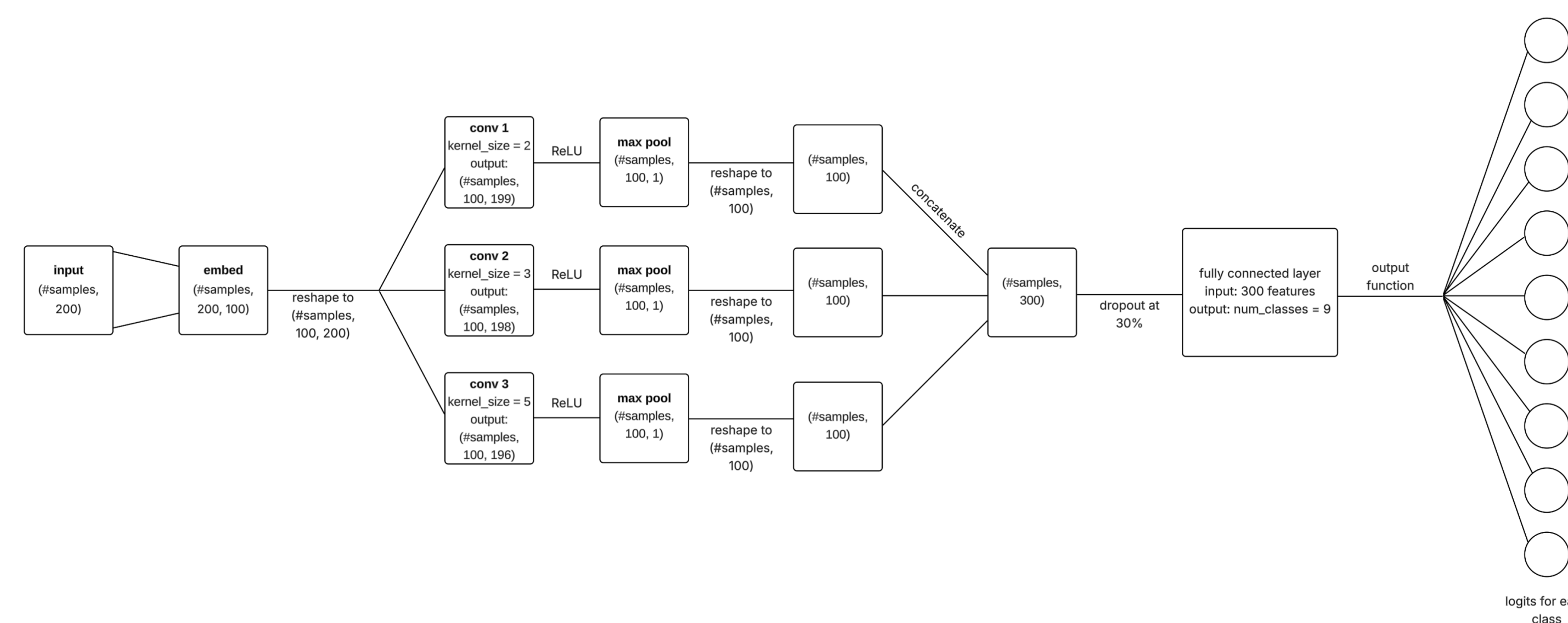
Results: Bayesian Approach

From the figure below, optimal results were found at n (num words) = 20 with a 50% Became Law Prior. These parameters yielded an overall accuracy of 57.85%, 44.44% accuracy on Bills that Became Law, and when the model predicts that a Bill becomes law it is right 96.55% of the time. The test dataset was completely unseen with 400 samples, optimal parameters chosen based on validation dataset of 400 samples.



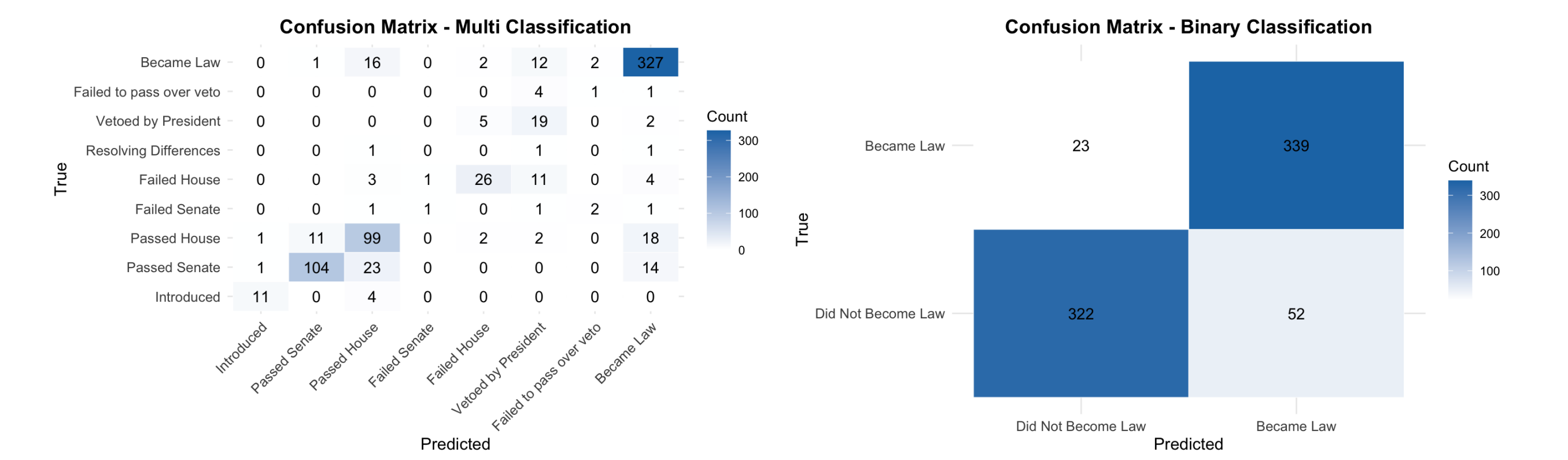
Methodology: CNN Approach

Two text classification CNNs were created with the same model architecture but different output and loss functions, a multi-class model and binary model. The input data was built by converting the bill text to vectors where each word is an element, then replaced with integer IDs representing each unique word. Through backpropagation, the embedding layer converts the word IDs to semantic meanings, allowing the CNN to learn relationships between words. Both models consist of the embedding layer, 3 parallel convolutional layers with kernel sizes of 2, 3, and 5, ReLU activation functions, max pooling, 30% dropout, and an output function that returns logits for each class (scores of how strongly the model thinks each class is correct). The different kernel sizes are for capturing different phrase lengths; a kernel size of 2 captures short 2-worded phrases while a kernel size of 5 captures longer phrases. The resulting data is concatenated to hold the most significant 2-gram, 3-gram, and 5-gram phrases in each bill text. The multi-class model uses the softmax output function and cross entropy loss for the loss function, while the binary classification model uses the sigmoid output function and binary cross entropy for the loss function.



Results: CNN Approach

The overall accuracy for the multi-class model on the test set was 79.89%, correctly classifying bills that became laws 74.43% of the time. The confusion matrix below demonstrates that the model correctly predicts most outcomes very well. It most commonly inaccurately predicts bills that became laws as 'Passed House', which makes sense because bills have to be approved by both chambers to be become law, so language that is consistent with bill passage would also likely be found in proposals approved by either chamber. The overall accuracy for the binary classification model was 89.8%, correctly classifying bills that became laws 93.6% of the time. The confusion matrix shows that the model incorrectly predicted that a bill became a law more frequently than it incorrectly classified bills that did not become laws.



Discussion and Future Work

The Bayesian Text Classification Model Results only predicted two outcomes, "Became Law" and "Introduced", but because it predicted all "Introduced" properly, this means that it still marks an improvement over guessing. Further work needs to be done to find the optimal prior that spreads out the current "Introduced" guesses into the other categories as well. This can be done by reducing the prior on "Introduced" while increasing other categories that were not predicted. This was partially tested with one of the 98% "Became Law" prior model, and helped boost the overall accuracy. However, because the other metrics regarding "Became Law" were not as strong, this prior was not chosen, but it provides a framework for a method to increase overall accuracy.

The binary classification CNN likely performs better than the multi-class model because it has a simpler decision boundary, it only has to decide if a bill belongs to 1 of 2 classes versus 1 of 9. Another improvement that could be made to these models is another parallel convolutional layer with a kernel size of one to capture specific words that contribute to bill outcome. The most significant n-grams for each outcome class could also be studied from these models which would give lawmakers more specific information about what will increase the chances of their proposed bill becoming a law.

References

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